

Microplastic Degradation Research Project

Phase 2 — Structure Analysis and Mutation Candidate Identification

Researcher: Arjun Popuri | Date: April 2026 | Script: step2_structure_analysis.py

This document records Phase 2: downloading PDB 5XJH, computing SASA for all 263 residues using the ShrakeRupley algorithm, identifying surface-exposed charge-reversal candidates, and finding disulfide bond candidates by Cbeta-Cbeta distance screening.

Key Concepts

What SASA measures

Solvent Accessible Surface Area measures how exposed each amino acid is to surrounding water. A relative SASA above 0.25 means surface-exposed — pepsin and acid can reach it. Below 0.25 means buried inside the protein core and protected.

The catalytic triad exclusion zone

Ser160, Asp206, and His237 form the catalytic triad that cuts PET. Any mutation within 8 Angstroms of these residues risks destroying enzyme activity. All mutation candidates must fall outside this exclusion zone.

Phase 2 Results

Structure downloaded and analyzed

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PDB: 5XJH (IsPETase, 1.5 Angstrom resolution, Joo et al. 2018)
Standard residues in chain A: 263
Surface-exposed (rel_ASA >= 0.25): 66 residues (25.1%)
Catalytic triad: Ser160 / Asp206 / His237 -- confirmed
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Charge-reversal candidates: 7 found, 5 selected

Pos	AA	Rel SASA	Dist to Triad	Proposed	Decision
53	R	0.2881	27.07	R53E	Excluded — near disulfide
59	R	0.6787	26.01	R59E	SELECTED — highest exposure
90	R	0.2529	12.14	R90E	Excluded — too close to triad
95	K	0.3964	14.71	K95E	Excluded — too close to triad
132	R	0.3978	22.29	R132E	SELECTED

224	R	0.5357	22.69	R224E	SELECTED
280	R	0.4755	15.01	R280E	SELECTED — subsite IIc
285	R	0.4000	15.01	R285E	SELECTED

Best disulfide candidate

Residue 1	Residue 2	Cb-Cb (Å)	Avg rel SASA	Min dist to triad	Decision
N233	S282	3.869	0.3590	10.35 Å	BEST CANDIDATE

Validation: The N233C/S282C pair independently identified here matches exactly the pair published by Zhong-Johnson et al. (2021), which raised IsPETase T_m to 69.4 degrees C. This confirms the structural analysis methodology is correct.

Summary

Finding	Value	Significance
Residues in crystal	263	Signal peptide absent from crystal structure
Surface-exposed residues	66 / 263	75% of protein is buried and protected
Charge-reversal candidates	7 found, 5 selected	Each R->E removes +2 charge units
Best disulfide pair	N233C + S282C	Cb-Cb = 3.869 Å, ideal geometry
Catalytic triad	Ser160 / Asp206 / His237	Confirmed, protected by 8 Å zone